A Dynamic Logic of the Right to Know

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Abstract

Different meanings of the "right to know" can be distinguished based on the theory of normative positions. In this paper, we focus on one of them—the power to know. Intuitively, in a sender-receiver setting, the receiver's power to know whether φ is the case means that the sender is obliged to (truthfully) announce the answer if the receiver asks the question φ?. Therefore, we develop a logic called LRK for reasoning about the power to know, the obligatory announcements, and the dynamics of questions and public announcements. We explore some semantic results for LRK and study the expressive power of fragments of LRK. In particular, we show that no DEL-style reduction axiomatization exists for LRK.

Keywords: epistemic rights, public announcement logic, logic of questions

1 Introduction

The aim of this paper is to provide a logical framework to reason about the notion of the right to know and its interaction with other related notions. As a theoretical-conceptual background, we rely on the theory of normative positions [18] based on the theory of Hohfeld [9,15]. Using this analysis, we can differentiate between four different meanings of the right to know: the privilege to know, the claim-right to know, the power to know, and the immunity to know. The difference between the four meanings of the right to know can be illustrated by the following example [23]: suppose that an agent i has been tested for some disease by his doctor. It is usually mentioned in laws that i has the right to know the test result. However, it is unclear which type of right they mean by it. i has a privilege-right to know the result which means that i has no duty not to know them, while i has a claim-right to know the result meaning that his doctor has a duty to inform him about the result. Besides,

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i has a power-right to know the result meaning that his doctor has a duty to inform them if i requests it. Last, i has an immunity-right to know the results, which protects him from his doctor taking away or altering his claim-right to know the results.

In the logical literature, so far the claim-right to know received explicit attention, see, e.g., [16]. In this paper, we will investigate the right to know as a power. Power is characterized in [15] as a potential: the agent having the power is able to execute an action resulting in the counterparty's normative positions changing, e.g. a duty arising. Thus in this case: the patient's asking about the test results creates the doctor's duty to tell.

The notion of the power to know can also be found extensively in database theory under the name of "the right/permission to access" (see, e.g., [5]). The logical characterization of "the right/permission to access" is crucial for practical problems like maintaining security in databases. The point of the problem is that the database has to answer the users' queries while complying with certain security policies (e.g., privacy policies). Many factors affecting the solutions to the problem have been identified in the literature. E.g., the representation of the database, the expressiveness of the query language, the space of admissible responses, the initial knowledge of the user, etc [3]. However, the expressiveness of the language for specifying security policies has somehow been overlooked. To the best of our knowledge, all existing works on the topic consider the permitted, forbidden, and obligatory knowledge/belief of users as the only components of a security policy, see, e.g., [3,5,1]. However, as a counter-example, it is stated in the General Data Protection Regulation of Europe (GDPR) that "A data subject should have *the right of access* to personal data which have been collected concerning him or her". We illustrate by a toy example the role of "the right/permission to access" in solving the security problem for databases. Consider two security policies $\mathcal{P}_1$ and $\mathcal{P}_2$ where $\mathcal{P}_1$ consists of the following clauses (1) and (2), and $\mathcal{P}_2$ the clauses (1), (2), and (3).

(1) The user i is permitted to know whether p.
(2) The user i is forbidden to know whether q.
(3) The user i has the right to access the answer to whether p.

Suppose further that the user i knows that p is equivalent to q. If the user i asks the database whether p holds, there are two different situations. Under the security policy $\mathcal{P}_1$, the database has solutions to this query, e.g., the database may keep silent.[3] However, under the security policy $\mathcal{P}_2$, no solution exists because the database is forced to answer the query $p?$.

In this paper, we develop a Logic of the Right to Know, LRK, to reason about the notion of the power-type of right to know whether something is the case (we often will refer to it as a 'power to know' in what follows). As suggested

[3] A dispute may arise on whether the database may keep silent. This depends on our understanding of "permission to know": does the user's permission to know whether p imply that the database needs to answer the query $p?$ raised by the user? But this is exactly a sign that we need different formalizations for these two different notions.

by the previous examples, the notion of the power to know is closely intertwined with other notions such as the obligation to inform, the public announcement, and the dynamics of questions. Therefore, LRK is devised such that these notions can also be expressed in the language.

The paper is structured as follows. In the next section, we introduce the language and semantics of LRK, as well as an example illustrating them. Section 3 and Section 4 are devoted to some semantic results of LRK and the expressive power of fragments of LRK, respectively. We discuss related literature in Section 5 and conclude with Section 6.

2 Language and Semantics

In this section, we introduce the language and semantics of LRK and illustrate them by an example. The scenarios that LRK is intended to characterize are communications between two agents where the information can only be transmitted from one agent (the sender/speaker, indicated by s) to the other (the receiver/addressee, indicated by r).The sender is further subject to some security policies such as privacy policies. These scenarios are common in our life, e.g., communications between a database and its users, conversations between a doctor and their patients, etc. We fix the role of the sender making only one of the agents able to make announcements. The restriction may seem to be unnatural. But, for simplicity, we will only consider the restricted scenarios. The following will serve as the running example of this paper:

Example 2.1 Two patients a and b have been tested for some diseases by the same doctor in the same time period. The policy consists of the following clauses: (1) The doctor is obliged to inform the patients about their test results; (2) Any patient is forbidden to know others' results; (3) The patients have the power to know whether the cheaper medicine is as good as the expensive one.

Suppose that, unfortunately, the test results for both a and b are positive and the cheaper medicine has the same efficacy as the expensive one. Let the sender be the doctor and the receiver the patient a. The doctor needs to decide which information should be informed.

Let us first introduce the formal language. Let PROP be a countable infinite set of propositional variables.

Definition 2.2 The language $\mathcal{L}$ is given by the following BNF grammar:

$$\varphi ::= p \mid \neg\varphi \mid (\varphi \to \varphi) \mid K_r\varphi \mid R_r\varphi \mid \mathbb{O}_s\varphi \mid \Box\varphi \mid [r\,{:}\,\varphi?]\varphi \mid [s\,{:}\,\varphi!]\varphi$$

where $p \in$ PROP. Other boolean connectives are defined as usual and $\langle s\,{:}\,\varphi!\rangle\psi$ is an abbreviation for $\neg[s\,{:}\,\varphi!]\neg\psi$.

The readings of the operators in $\mathcal{L}$ are as follows:

- $K_r\varphi$: The receiver knows φ.
- $R_r\varphi$: The receiver has the power-right to know the answer to the question φ? (or, equivalently, the receiver has the power-right to know whether φ).

- $\mathbb{O}_s\varphi$: The sender is obliged to (truthfully) announce φ.
- $\Box\varphi$: It is universally true that φ.
- $[r\!:\!\varphi?]\psi$: After the receiver asked the question $\varphi?$, it holds that ψ.
- $[s\!:\!\varphi!]\psi$: After the sender (truthfully) announced φ, it holds that ψ.

Next, we introduce the models for LRK. Our models are essentially a combination of the "neighbourhood epistemic models" introduced in [12] and the "epistemic issue models" in [21], see Section 5.

Definition 2.3 A model is a tuple $M = (W, \sim, \approx, N, V)$ where:

- W is a non-empty set of possible worlds or states;
- $\sim \subseteq W \times W$ is an equivalence relation on W;
- $\approx \subseteq \wp(W)$ is a partition of W;
- $N : W \to \wp(\wp(W))$ is such that $w \in U$ for all $w \in W$ and $U \in N(w)$;
- $V : \textsc{prop} \to \wp(W)$ is a valuation.

A pointed model is a pair M, w such that w is a state of M. For every state $w \in W$, we denote the set $\{v \in W \mid w \sim v\}$ as $\sim(w)$. We will also denote the unique subset in $\approx$ that contains w as $\approx(w)$.

In the above definition, $\sim$ is the familiar epistemic indistinguishability relation (of the receiver). The partition $\approx$ is intended to encode the set of questions to which the receiver has the power to know the answers. The set of questions may be stipulated by some normative systems such as privacy policies. The idea to represent a set of questions by a partition can be found in, e.g., [7], [4], and [21]. Finally, each subset in the neighbourhood $N(w)$ is an ideal epistemic state for the receiver at w. Thus the neighbourhood function N specifies which epistemic states (of the receiver) are ideal at every state w.

Let us illustrate the definition of the models by the running example:

Example 2.4 Let p_a, p_b, and g be the propositions that "The result for a is positive", "The result for b is positive", and "The cheaper medicine is as good as the expensive one", respectively. The case in Example 2.1 can be characterized by the model $M = (W, \sim, \approx, N, V)$ (as illustrated in Fig. 1) where:

- $W = \{000, 001, 010, 011, 100, 101, 110, 111\}$;
- $\sim = W \times W$;
- $\approx = \{\{000, 010, 100, 110\}, \{001, 011, 101, 111\}\}$;
- for all $xyz \in \{000, 001, 010, 011\}$, $N(xyz) = \{U \mid xyz \in U \;\&\; U \subseteq \{000, 001, 010, 011\} \;\&\; U \not\subseteq \{000, 001\} \;\&\; U \not\subseteq \{010, 011\}\}$,
  for all $xyz \in \{100, 101, 110, 111\}$, $N(xyz) = \{U \mid xyz \in U \;\&\; U \subseteq \{100, 101, 110, 111\} \;\&\; U \not\subseteq \{100, 101\} \;\&\; U \not\subseteq \{110, 111\}\}$;
- $V(p_a) = \{xyz \in W \mid x = 1\}$,
  $V(p_b) = \{xyz \in W \mid y = 1\}$,
  $V(g) = \{xyz \in W \mid z = 1\}$.

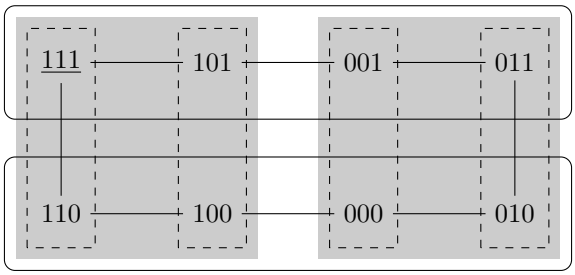

Fig. 1. The model $M = (W, \sim, \approx, N, V)$. States are 000, 001, ... and the actual state is 111. For each state xyz, $x = 1$ ($y = 1$, $z = 1$, respectively) iff $xyz \in V(p_a)$ ($xyz \in V(p_b)$, $xyz \in V(g)$, respectively). The indistinguishability relation $\sim$ is indicated by the straight line (with the reflexive and transitive arrows omitted). $\approx$ is the partition indicated by the rectangles with rounded corners (this captures the patient's power to know g). Finally, for every $xyz \in W$, $N(xyz)$ consists of the subsets such that: (1) it contains xyz itself; (2) it is contained in one of the shaded areas (this corresponds to the doctor's obligation to inform about p_a); (3) it is not contained in one of the dashed rectangles (the patient is prohibited to know p_b).

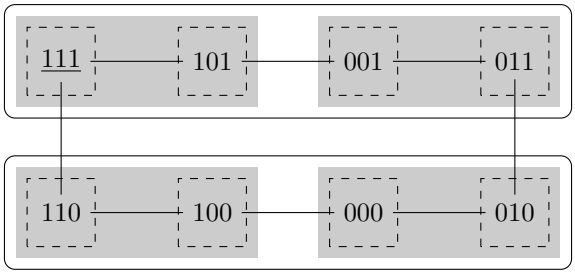

Fig. 2. The updated model $M_{g?}$. The same convention is adopted as in Fig. 1.

The next step is to provide the semantics for $\mathcal{L}$, especially for the formulas $R_r\varphi$ and $[r : \varphi?]\psi$. The semantics for $R_r\varphi$ is relatively straightforward: the receiver has the power to know the answer to φ? iff the question φ? is "settled" by the partition $\approx$, in the sense that all the cells in the partition $\approx$ are subsets of the truth set of φ or $\neg\varphi$. One may already notice that the truth of $R_r\varphi$ does not depend on the evaluating states. Hence, the notion of the power to know is a global notion in our models.[4] To express this fact, we use the universal modality $\Box$.

For the semantics for $[r : \varphi?]\psi$, as mentioned before, $[r : \varphi?]\psi$ is intended to express that "after the receiver asked the question φ?, it holds that ψ". If the receiver has no power to know the answer to φ?, then nothing will change after the action $[r : \varphi?]$. On the contrary, if the receiver indeed has the power, the sender is then forced to answer the question φ?. This is modeled in our

[4] It also makes sense to generalize our models to account for cases where, for example, the receiver is not aware of their rights to know. We leave that for future work.

framework in a way such that, in the updated model, all epistemic states that are not answers to $\varphi?$ become no longer ideal. Formally, the idea is captured by the definition of $M_{\varphi?}$ in the next definition.

Definition 2.5 Given a model $M = (W, \sim, \approx, N, V)$, for all $w \in W$ and $\varphi \in \mathcal{L}$, the satisfaction relation $M, w \models \varphi$ is inductively defined as follows:

$$\begin{array}{rcl}
M, w \models p & \text{iff} & w \in V(p) \\
M, w \models \neg\varphi & \text{iff} & M, w \not\models \varphi \\
M, w \models (\varphi \to \psi) & \text{iff} & M, w \not\models \varphi \text{ or } M, w \models \psi \\
M, w \models K_r\varphi & \text{iff} & \text{for all } v \in W,\ w \sim v \text{ implies } M, v \models \varphi \\
M, w \models R_r\varphi & \text{iff} & \text{for all } U \in \approx,\ U \subseteq [\![\varphi]\!]_M \text{ or } U \subseteq [\![\neg\varphi]\!]_M \\
M, w \models \mathbb{O}_s\varphi & \text{iff} & \text{for all } U \in N(w),\ U \subseteq \sim(w) \text{ implies } U \subseteq [\![\varphi]\!]_M \\
M, w \models \Box\varphi & \text{iff} & \text{for all } v \in W,\ M, v \models \varphi \\
M, w \models [r\!:\!\varphi?]\psi & \text{iff} & M_{\varphi?}, w \models \psi \\
M, w \models [s\!:\!\varphi!]\psi & \text{iff} & M, w \models \varphi \text{ implies } M_{\varphi!}, w \models \psi
\end{array}$$

where $[\![\varphi]\!]_M = \{x \in W \mid M, x \models \varphi\}$ is the truth set of φ in M and the models $M_{\varphi?}$ and $M_{\varphi!}$ are defined as follows:

$M_{\varphi?} = M$ if $M, w \not\models R_r\varphi$; otherwise $M_{\varphi?} = (W_{\varphi?}, \sim_{\varphi?}, \approx_{\varphi?}, N_{\varphi?}, V_{\varphi?})$ where:

- $W_{\varphi?} = W$, $\sim_{\varphi?} = \sim$, $\approx_{\varphi?} = \approx$, $V_{\varphi?} = V$,
- $N_{\varphi?}(x) = \{U \in N(x) \mid U \subseteq [\![\varphi]\!]_M \text{ or } U \subseteq [\![\neg\varphi]\!]_M\}$ for all $x \in W$.

$M_{\varphi!} = (W_{\varphi!}, \sim_{\varphi!}, \approx_{\varphi!}, N_{\varphi!}, V_{\varphi!})$ where:

- $W_{\varphi!} = W$, $\approx_{\varphi!} = \approx$, $N_{\varphi!} = N$, $V_{\varphi!} = V$,
- $\sim_{\varphi!} = \sim \cap \{(u, v) \in W \times W \mid M, u \models \varphi \text{ iff } M, v \models \varphi\}$.

The semantics for $K_r\varphi$ and $[s\!:\!\varphi!]\psi$ is standard, except that in the definition of $M_{\varphi!}$ we choose to delete the links between the φ-states and $\neg\varphi$-states, instead of removing all $\neg\varphi$-states from the model. Those $\neg\varphi$-states are reserved for further reference.[5] This kind of model updating can be found in, e.g., [20,21]. The intuition behind the semantics for $R_r\varphi$ and $[r\!:\!\varphi?]\psi$ has been explained. As for $\mathbb{O}_s\varphi$, it reflects the intuition that the sender is obliged to announce φ if φ holds in all ideal epistemic states (for the receiver) that are achievable by further announcements (i.e., φ is a piece of necessary information for restoring ideality). Let us illustrate the semantics by the running example:

Example 2.6 In the model M, we have, e.g., $M, 111 \models \mathbb{O}_s p_a$, $M, 111 \models R_r g$, and $M, 111 \not\models \mathbb{O}_s p_b$. After the receiver (the patient a) asked "Is the cheaper medicine as good as the expensive one?" ($g?$), the updated pointed model is $M_{g?}$ in Figure 2. We have, e.g., $M_{g?}, 111 \models \mathbb{O}_s p_a$, $M_{g?}, 111 \models R_r g$, and $M_{g?}, 111 \models \mathbb{O}_s g$.

A question may arise regarding the semantics for $\mathbb{O}_s\varphi$: why do we just consider ideal epistemic states achievable by further announcements instead of

[5] This deviates from the classical public announcement logic. But, in our case, deleting all $\neg\varphi$-states may change the truth of formulas like $R_r p$, which is unreasonable.

all? Technically, one may propose the following alternative semantic definition for $\mathbb{O}_s\varphi$:

$$M, w \models \mathbb{O}_s\varphi \quad \text{iff} \quad \text{for all } U \in N(w),\ U \subseteq [\![\varphi]\!]_M. \tag{$\dagger$}$$

We will argue, however, that ($\dagger$) does not work in certain cases within the context of database security, whereas our original semantics does. Consider the following example taken from [1]:

Example 2.7 Suppose the sender is communicating classified information to the receiver. Since the receiver is permitted to know some information p, the only constraint for the sender is that it is forbidden for the receiver to know p while not to know that it is classified (c). Furthermore, suppose that the receiver currently knows p, but she does not know c. The scenario can be represented by the following model $M = (W, \sim, \approx, N, V)$ such that:

- $W = \{w, u, v, x\}$;
- $\sim(w) = \sim(u) = \{w, u\}$, $\sim(v) = \{v\}$, and $\sim(x) = \{x\}$;
- $\approx = \{W\}$;
- $N(w) = \{Y \subseteq W \mid w \in Y \text{ and } Y \neq \{w, u\}\}$;
- $V(p) = \{w, u\}$ and $V(c) = \{w, v\}$.

Intuitively, we will agree that, in this case, it is obligatory for the sender to announce c. This is consistent with our semantics since $M, w \models \mathbb{O}_s c$. However, it would not be the case if we adopted ($\dagger$).

When defining our models, the set of ideal epistemic states for a state w, $N(w)$, is not required to be non-empty. This is in contrast with standard deontic logic. Formally, we say a model $M = (W, \sim, \approx, N, V)$ is *standard* if for all $w \in W$, $N(w) \neq \varnothing$. The next proposition shows that the class of standard models gives the same logic as the class of all models.

Proposition 2.8 *For all pointed models M, w, there is a standard model M' and w' in M' such that $M, w \models \varphi$ iff $M', w' \models \varphi$ for all $\varphi \in \mathcal{L}$.*

Proof. Given a model $M = (W, \sim, \approx, N, V)$, we define the *double model* of M, $M' = (W', \sim', \approx', N', V')$, as follows:

- $W' = W \times \{1, 2\}$ (elements of W' will be denoted by $w_1, w_2, \dots$);
- $w_i \sim' v_j$ iff $w \sim v$ and $i = j$;
- $\approx' = \{U \times \{1\}, U \times \{2\} \mid U \in \approx\}$;
- for all $w_i \in W'$, $N'(w_i) = \{U \times \{i\} \mid U \in N(w)\} \cup \{\{w\} \times \{1, 2\}\}$;
- $V'(p) = V(p) \times \{1, 2\}$.

It is clear that $N'(w_i) \neq \varnothing$ for any $w_i \in W'$, hence M' is a standard model. Then the proposition follows from the following claim:

Claim. For all $w_i \in W'$, $M', w_i \models \varphi$ iff $M, w \models \varphi$ for all $\varphi \in \mathcal{L}$.
Proof of Claim. Induction on the structure of φ. Here we show only the inductive step for $\mathbb{O}_s\psi$: From left to right. Suppose $M', w_i \models \mathbb{O}_s\psi$. Let $U \in N(w)$ be such that $U \subseteq \sim(w)$. It suffices to show that $U \subseteq [\![\psi]\!]_M$. Note

that $U \times \{i\} \in N'(w_i)$ and $U \times \{i\} \subseteq \sim' (w_i)$ by the definition of N' and $\sim'$, respectively. Hence $U \times \{i\} \subseteq [\![\psi]\!]_{M'}$ by the assumption. It follows that $U \subseteq [\![\psi]\!]_M$ by the IH. From right to left. Suppose $M, w \models \mathbb{O}_s\psi$. Let $U' \in N'(w_i)$ be such that $U' \subseteq \sim' (w_i)$. It suffices to show that $U' \subseteq [\![\psi]\!]_{M'}$. Since $\{w\} \times \{1, 2\} \not\subseteq \sim' (w_i)$, there must be $U \in N(w)$ such that $U' = U \times \{i\}$. Note also that $U \subseteq \sim (w)$ since $U' \subseteq \sim' (w_i)$. Hence $U \subseteq [\![\psi]\!]_M$ by the assumption. Thus $U' \subseteq [\![\psi]\!]_{M'}$ by the IH.

For the case $[r : \psi?]\chi$ (and, similarly, $[s : \psi!]\chi$), note that $M'_{\psi?}$ is still the double model of $M_{\psi?}$ by the IH. Hence, by applying the IH, we have $M'_{\psi?}, w_i \models \chi$ iff $M_{\psi?}, w \models \chi$. □

3 Some Semantic Results

In this section, we list some (in)validities of LRK. The first group of validities is about the notion of the power to know. We omit the proofs because they are all straightforward.

Proposition 3.1 *The following hold for all formulas φ and ψ:*

(1) $\models R_r\top$.
(2) $\models R_r\varphi \wedge R_r\psi \to R_r(\varphi \wedge \psi)$.
(3) $\models R_r\varphi \to R_r\neg\varphi$.
(4) $\models \Box(\varphi \leftrightarrow \psi) \to (R\varphi \leftrightarrow R\psi)$.

From the above, we can see that the fragment of LRK on the notion of the power to know is nothing but the logical entailment relation between questions. That is to say, in LRK, if the answer to a question φ? can be derived from that of a set of questions that the receiver has the power to know the answers to, then the receiver also has the power to know the answer to φ?. [6]

Proposition 3.2 *The following hold for all formulas φ and ψ:*

(1) $\models K_r(\varphi \to \psi) \to (\mathbb{O}_s\varphi \to \mathbb{O}_s\psi)$.
(2) $\models \neg\mathbb{O}_s\bot \to (\mathbb{O}_s\varphi \to \varphi)$.
(3) $\models \mathbb{O}_s(\varphi \to \psi) \to (\mathbb{O}_s\varphi \to \mathbb{O}_s\psi)$.
(4) $\models \Box\varphi \to \mathbb{O}_s\varphi$.
(5) $\models K_r\varphi \to \mathbb{O}_s\varphi$.

The proofs are again omitted due to the same reason. A few remarks can be made on the above logical rules governing the behavior of the operator $\mathbb{O}_s$. The first says that if φ is more informative than ψ for the receiver and the sender is obliged to inform φ, [7] then the sender is also obliged to inform ψ.

[6] One may argue that the notion of the power to know characterized in LRK is a rather weak notion. For example, in most scenarios, the receiver has only the power to know what they are permitted to know (otherwise there would be conflict in the security policy). To model these scenarios, we can impose extra constraints on the models, e.g., for all $w \in W$, there is $U \in N(w)$ such that $\approx (w) \subseteq U$. Since the focus of this paper is on the formalization of the power to know, we leave this for future work.

[7] We follow [1, Definition 9] for the definition of "informativeness" of formulas.

The second says that the sender is not obliged to lie unless they are obliged to announce the contradiction (i.e., they face a deontic dilemma). Further, the validities (3) and (4) indicate that $\mathbb{O}_s\varphi$ is a normal modality. But we have a problem with interpreting the last validity. Literally, it states that the sender is obliged to announce whatever the receiver knows. This seems rather counterintuitive. To understand (5), note that, if φ is known by the receiver, the announcement of φ is actually less informative than any announcement for the receiver. Thus, from the informational point of view, the announcement of φ is "implied" by any announcement. In this sense, the announcement of φ is inevitable or necessary in our system since we assume that the sender can only make (truthful) announcements. So, the obligatory announcement of φ simply follows from the fact that the announcement of φ is necessary. [8]

The previous two propositions are about the properties of "the power to know" and "obligatory announcements" separately. However, it is natural to expect that there would be some interaction between them. One candidate is the formula $R_r\varphi \to (\varphi \to [r:\varphi?]\mathbb{O}_s\varphi)$, expressing that if the receiver has the power to know whether φ and φ is the case, then the sender is obliged to announce φ once the receiver has asked the question $\varphi?$. It is not hard to show the validity of the formula when φ is propositional. But the next proposition shows that this needs not to be the case when φ is a general formula:

Proposition 3.3 $\not\models R_r\varphi \to (\varphi \to [r:\varphi?]\mathbb{O}_s\varphi)$ *for some formulas* φ.

Proof. We show that $\not\models R_r(p\wedge\neg\mathbb{O}_sp) \to ((p\wedge\neg\mathbb{O}_sp) \to [r:(p\wedge\neg\mathbb{O}_sp)?]\mathbb{O}_s(p\wedge\neg\mathbb{O}_sp))$. Let the model $M = (W, \sim, \approx, N, V)$ be as follows:

- $W = \{w, v\}$, $\sim = W \times W$, $\approx = \{\{v\}, \{w\}\}$, $V(p) = \{w\}$,
- $N(w) = \{\{w\}, \{v, w\}\}$ and $N(v) = \{v, w\}$.

We are going to show that $M, w \not\models R_r(p \wedge \neg\mathbb{O}_sp) \to ((p \wedge \neg\mathbb{O}_sp) \to [r : (p \wedge \neg\mathbb{O}_sp)?]\mathbb{O}_s(p \wedge \neg\mathbb{O}_sp))$. First, it is not hard to see that $M, w \models p \wedge \neg\mathbb{O}_sp$ (1) and $M, v \not\models p\wedge\neg\mathbb{O}_sp$. Hence $[\![p\wedge\neg\mathbb{O}_sp]\!]_M = \{w\}$ and $[\![\neg(p\wedge\neg\mathbb{O}_sp)]\!]_M = \{v\}$. Therefore $M, w \models R_r(p\wedge\neg\mathbb{O}_sp)$ (2). In the updated model $M_{(p\wedge\neg\mathbb{O}_sp)?}$, the only change is that $N_{(p\wedge\neg\mathbb{O}_sp)?}(w) = \{\{w\}\}$. It follows that $M_{(p\wedge\neg\mathbb{O}_sp)?}, w \models \mathbb{O}_sp$. Thus $M_{(p\wedge\neg\mathbb{O}_sp)?}, w \not\models \mathbb{O}_s(p\wedge\neg\mathbb{O}_sp)$. Therefore $M, w \not\models [r:(p\wedge\neg\mathbb{O}_sp)?]\mathbb{O}_s(p\wedge\neg\mathbb{O}_sp)$ (3). By (1), (2), and (3), $M, w \not\models R_r(p \wedge \neg\mathbb{O}_sp) \to ((p \wedge \neg\mathbb{O}_sp) \to [r : (p \wedge \neg\mathbb{O}_sp)?]\mathbb{O}_s(p \wedge \neg\mathbb{O}_sp))$. □

In the above proof, the formula $p\wedge\neg\mathbb{O}_sp$ is used to show the invalidity of the given axiom schema. The formula has the same structure as the Moore sentence [17], i.e., p is true but I do not believe p. It is well known, in dynamic epistemic logic [22], that the Moore sentence $(p \wedge \neg Kp)$ is an unsuccessful formula, in the sense that the Moore sentence may become false after the announcement of itself. Here we see a similar situation: after the receiver asks the question $p\wedge\neg\mathbb{O}_sp?$, it becomes obligatory for the sender to announce p. Thus the truth of the formula $p \wedge \neg\mathbb{O}_sp$ becomes false. However, the operator $\mathbb{O}_s$ satisfies a

[8] This can be seen as an analogy to the rule of necessitation in SDL.

weak form of the axiom (T): $\models \neg\mathbb{O}_s\bot \to (\mathbb{O}_s\varphi \to \varphi)$ (Proposition 3.2(2)). Hence, in the updated model, it is not the case that $\mathbb{O}_s(p \wedge \neg\mathbb{O}_s p)$.

At first glance, the invalidity of the axiom schema in Proposition 3.3 may seem to be counterintuitive. How could it be that the receiver has the power to know something while the sender has no obligation to inform even if the receiver requests it? We will, nevertheless, argue that the phenomenon can be explained if we make explicit the time involved in the axiom schema. The operator $\mathbb{O}_s\varphi$ in LRK expresses veritable obligations ([8], i.e., obligations specific to a particular situation) instead of normative rules. This means that the truth of formulas like $\mathbb{O}_s\varphi$ may flip after the receiver asks some questions because the situation changes. Consider the formula $R_r(p \wedge \neg\mathbb{O}_s p) \to ((p \wedge \neg\mathbb{O}_s p) \to [r : (p\wedge\neg\mathbb{O}_s p)?]\mathbb{O}_s(p\wedge\neg\mathbb{O}_s p))$. The first three occurrences of the operator $\mathbb{O}_s$ really refer to the obligation of the sender *before* the question $(p \wedge \neg\mathbb{O}_s p)?$, whereas the last two occurrences express the sender's obligation *after* the question. Thus, the antecedent $R_r(p\wedge\neg\mathbb{O}_s p)$ just asserts that the receiver has the power to know the sender's deontic status before the question. However, in LRK, there is no way to express the sender's previous obligation in the scope of the dynamic operator $[r : p \wedge \neg\mathbb{O}_s p?]$. This suggests that LRK may be equipped with temporal operators like "Yesterday".

The next two propositions are on the behavior of the two dynamic operators $[s\!:\!\varphi!]$ and $[r\!:\!\varphi?]$. Like in dynamic epistemic logic, a series of reduction axioms exist for these two operators. In order to show this, we need Lemma 3.4.

Lemma 3.4 *The following hold:*

(1) for any model M and formulas φ, ψ, $[\![\langle s\!:\!\varphi!\rangle\psi \vee \langle s\!:\!\neg\varphi!\rangle\psi]\!]_M = [\![\psi]\!]_{M_{\varphi!}}$.

(2) for any pointed model M, w and formulas φ, ψ, if $M, w \models R_r\varphi$ then $[\![[r\!:\!\varphi?]\psi]\!]_M = [\![\psi]\!]_{M_{\varphi?}}$.

Proof. The proof for (2) is straightforward. We consider only (1): let $M = (W, \sim, \approx, N, V)$ and $w \in W$. We have

$$\begin{array}{lll}
 & M, w \models \langle s\!:\!\varphi!\rangle\psi \vee \langle s\!:\!\neg\varphi!\rangle\psi & \\
\text{iff} & M, w \models \langle s\!:\!\varphi!\rangle\psi \text{ or } M, w \models \langle s\!:\!\neg\varphi!\rangle\psi & \text{(semantics)} \\
\text{iff} & (M, w \models \varphi \text{ and } M_{\varphi!}, w \models \psi) \text{ or} & \text{(semantics)} \\
 & (M, w \models \neg\varphi \text{ and } M_{\neg\varphi!}, w \models \psi) & \\
\text{iff} & (M, w \models \varphi \text{ and } M_{\varphi!}, w \models \psi) \text{ or} & (M_{\varphi!} = M_{\neg\varphi!}) \\
 & (M, w \models \neg\varphi \text{ and } M_{\varphi!}, w \models \psi) & \\
\text{iff} & M_{\varphi!}, w \models \psi &
\end{array}$$

Proposition 3.5 *The following hold for all formulas φ and ψ:*

(1) $\models [s\!:\!\varphi!]p \leftrightarrow (\varphi \to p)$.

(2) $\models [s\!:\!\varphi!]\neg\psi \leftrightarrow (\varphi \to \neg[s\!:\!\varphi!]\psi)$.

(3) $\models [s\!:\!\varphi!](\psi \to \chi) \leftrightarrow ([s\!:\!\varphi!]\psi \to [s\!:\!\varphi!]\chi)$.

(4) $\models [s\!:\!\varphi!]K_r\psi \leftrightarrow (\varphi \to K_r[s\!:\!\varphi!]\psi)$.

(5) $\models [s\!:\!\varphi!]R_r\psi \leftrightarrow (\varphi \to R_r(\langle s\!:\!\varphi!\rangle\psi \vee \langle s\!:\!\neg\varphi!\rangle\psi))$.

(6) $\models [s\!:\!\varphi!]\Box\psi \leftrightarrow (\varphi \to \Box(\langle s\!:\!\varphi!\rangle\psi \vee \langle s\!:\!\neg\varphi!\rangle\psi))$.

(7) if $\models \varphi$ *then* $\models [s\!:\!\psi!]\varphi$.

Proof. We show only (5). Let $M = (W, \sim, \approx, N, V)$ and $w \in W$. We have

$$\begin{array}{lll}
 & M, w \models [s\!:\!\varphi!]R_r\psi & \\
\text{iff} & M, w \models \varphi \text{ implies } M_{\varphi!}, w \models R_r\psi & \text{(semantics)} \\
\text{iff} & M, w \models \varphi \text{ implies} & \text{(semantics)} \\
 & (\forall U \in \approx_{\varphi!}\text{: } U \subseteq [\![\psi]\!]_{M_{\varphi!}} \text{ or } U \subseteq [\![\neg\psi]\!]_{M_{\varphi!}}) & \\
\text{iff} & M, w \models \varphi \text{ implies} & \text{(def. of } M_{\varphi!}) \\
 & (\forall U \in \approx\text{: } U \subseteq [\![\psi]\!]_{M_{\varphi!}} \text{ or } U \subseteq [\![\neg\psi]\!]_{M_{\varphi!}}) & \\
\text{iff} & M, w \models \varphi \text{ implies} & \text{(Lemma 3.4(1))} \\
 & (\forall U \in \approx\text{: } (U \subseteq [\![\langle s\!:\!\varphi!\rangle\psi \vee \langle s\!:\!\neg\varphi!\rangle\psi]\!]_M \text{ or} & \\
 & U \subseteq [\![\neg(\langle s\!:\!\varphi!\rangle\psi \vee \langle s\!:\!\neg\varphi!\rangle\psi)]\!]_M)) & \\
\text{iff} & M, w \models \varphi \text{ implies} & \text{(semantics)} \\
 & M, w \models R_r(\langle s\!:\!\varphi!\rangle\psi \vee \langle s\!:\!\neg\varphi!\rangle\psi) & \\
\text{iff} & M, w \models \varphi \to R_r(\langle s\!:\!\varphi!\rangle\psi \vee \langle s\!:\!\neg\varphi!\rangle\psi) & \text{(semantics)}
\end{array}$$

Proposition 3.6 *The following hold for all formulas* φ *and* ψ*:*

(1) $\models [r\!:\!\varphi?]p \leftrightarrow p$.
(2) $\models [r\!:\!\varphi?]\neg\psi \leftrightarrow ((\neg R_r\varphi \wedge \neg\psi) \vee (R_r\varphi \wedge \neg[r\!:\!\varphi?]\psi))$.
(3) $\models [r\!:\!\varphi?](\psi \to \chi) \leftrightarrow ([r\!:\!\varphi?]\psi \to [r\!:\!\varphi?]\chi)$.
(4) $\models [r\!:\!\varphi?]K_r\psi \leftrightarrow ((K_r\psi \wedge \neg R_r\varphi) \vee (K_r[r\!:\!\varphi?]\psi \wedge R_r\varphi))$.
(5) $\models [r\!:\!\varphi?]R_r\psi \leftrightarrow ((R_r\psi \wedge \neg R_r\varphi) \vee (R_r[r\!:\!\varphi?]\psi \wedge R_r\varphi))$.
(6) $\models [r\!:\!\varphi?]\Box\psi \leftrightarrow ((\Box\psi \wedge \neg R_r\varphi) \vee (\Box[r\!:\!\varphi?]\psi \wedge R_r\varphi))$.

Proof. We show only (2) and (4). Let $M = (W, \sim, \approx, N, V)$ and $w \in W$. (2): It is clear that $M, w \models [r\!:\!\varphi?]\neg\psi \wedge \neg R_r\varphi$ iff $M, w \models \neg\psi \wedge \neg R_r\varphi$ $(*)$. On the other hand, we have:

$$\begin{array}{lll}
 & M, w \models [r\!:\!\varphi?]\neg\psi \wedge R_r\varphi & \\
\text{iff} & M_{\varphi?}, w \models \neg\psi \text{ and } M, w \models R_r\varphi & \text{(semantics)} \\
\text{iff} & M_{\varphi?}, w \not\models \psi \text{ and } M, w \models R_r\varphi & \text{(semantics)} \\
\text{iff} & M, w \not\models [r\!:\!\varphi?]\psi \text{ and } M, w \models R_r\varphi & \text{(semantics)} \\
\text{iff} & M, w \models \neg[r\!:\!\varphi?]\psi \wedge R_r\varphi & \text{(semantics)}
\end{array}$$

Hence $\models ([r:\varphi?]\neg\psi \wedge R_r\varphi) \leftrightarrow (\neg[r:\varphi?]\psi \wedge R_r\varphi)$ $(**)$. From $(*)$ and $(**)$ it follows that $\models [r\!:\!\varphi?]\neg\psi \leftrightarrow ((\neg R_r\varphi \wedge \neg\psi) \vee (R_r\varphi \wedge \neg[r\!:\!\varphi?]\psi))$.

(4): It is clear that $\models ([r:\varphi?]K_r\psi \wedge \neg R_r\varphi) \leftrightarrow (K_r\psi \wedge \neg R_r\varphi)$ $(*)$. On the other hand:

$$\begin{array}{ll}
 & M, w \models [r\!:\!\varphi?]K_r\psi \wedge R_r\varphi \\
\text{iff} & M_{\varphi?}, w \models K_r\psi \text{ and } M, w \models R_r\varphi \text{ (semantics)} \\
\text{iff} & (\forall v \in W_{\varphi?}\text{: } w \sim_{\varphi?} v \text{ implies } M_{\varphi?}, v \models \psi) \text{ and } M, w \models R_r\varphi \text{ (semantics)} \\
\text{iff} & (\forall v \in W\text{: } w \sim v \text{ implies } M_{\varphi?}, v \models \psi) \text{ and} M, w \models R_r\varphi \text{ (def. of } M_{\varphi?}) \\
\text{iff} & (\forall v \in W\text{: } w \sim v \text{ implies } M, v \models [r\!:\!\varphi?]\psi) \text{ and } M, w \models R_r\varphi \\
 & \text{(Lemma 3.4(2))} \\
\text{iff} & M, w \models K_r[r\!:\!\varphi?]\psi \wedge R_r\varphi \text{ (semantics)}
\end{array}$$

Therefore $\models ([r:\varphi?]K_r\psi \wedge R_r\varphi) \leftrightarrow (K_r[r:\varphi?]\psi \wedge R_r\varphi)$ $(**)$. From $(*)$ and

(∗∗) it follows that $\models [r:\varphi?]K_r\psi \leftrightarrow ((K_r\psi \wedge \neg R_r\varphi) \vee (K_r[r:\varphi?]\psi \wedge R_r\varphi))$. □

4 Expressivity

In this section, we investigate the expressive power of some fragments of $\mathcal{L}$. As we have seen, there are a couple of operators in $\mathcal{L}$. But it is not clear whether some of them are superfluous. In particular, we are going to address the following two questions in the section:

- Does adding the two dynamic operator $[s:\varphi!]$ and $[r:\varphi?]$ to $\mathcal{L}$ increases the expressive power?
- Is the operator $R_r\varphi$ for the power to know expressible in a language without it?

It is well known that adding the public announcement operator to the language of epistemic logic (without common knowledge) does not increase expressive power. In Section 3, we have seen that there exists a series of reduction axioms for the two dynamic operators $[s:\varphi!]$ and $[r:\varphi?]$, but not for formulas of the form $[s:\varphi!]\mathbb{O}_s\psi$ and $[r:\varphi?]\mathbb{O}_s\psi$. Thus it is interesting to know whether the same holds for LRK. To answer the first question, let $\mathcal{L}_0$, $\mathcal{L}_1$, and $\mathcal{L}_2$ be the sublanguages of $\mathcal{L}$ defined as follows:

$$\varphi ::= p \mid \neg\varphi \mid (\varphi \to \varphi) \mid K_r\varphi \mid R_r\varphi \mid \mathbb{O}_s\varphi \mid \Box\varphi \qquad (\mathcal{L}_0)$$
$$\varphi ::= p \mid \neg\varphi \mid (\varphi \to \varphi) \mid K_r\varphi \mid R_r\varphi \mid \mathbb{O}_s\varphi \mid \Box\varphi \mid [s:\varphi!]\varphi \qquad (\mathcal{L}_1)$$
$$\varphi ::= p \mid \neg\varphi \mid (\varphi \to \varphi) \mid K_r\varphi \mid R_r\varphi \mid \mathbb{O}_s\varphi \mid \Box\varphi \mid [r:\varphi?]\varphi \qquad (\mathcal{L}_2)$$

The next theorem shows that adding any of the two operators $[s:\varphi!]\psi$ and $[r:\varphi?]\psi$ to the static language $\mathcal{L}_0$ does increase the expressive power. This is in contrast with the situation in public announcement logic and gives a positive answer to the first question. It also follows that no DEL-style reduction axiomatization exists for LRK.

Theorem 4.1 *The following hold:*

(1) $\mathcal{L}_1$ *is more expressive than* $\mathcal{L}_0$.

(2) $\mathcal{L}_2$ *is more expressive than* $\mathcal{L}_0$.

Proof. (1): It is clear that $\mathcal{L}_1$ is at least as expressive as $\mathcal{L}_0$ since $\mathcal{L}_0$ is a sublanguage of $\mathcal{L}_1$. We show that $\mathcal{L}_0$ is not at least as expressive as $\mathcal{L}_1$. This is done by showing that there is no $\psi \in \mathcal{L}_0$ such that $[s:p!]\mathbb{O}_s\bot \equiv \psi$ (i.e., they are satisfied at exactly the same pointed models). Consider two models $M_1 = (W, \sim, \approx, N_1, V)$ and $M_2 = (W, \sim, \approx, N_2, V)$ where:

- $W = \{w, u\}$, $\sim = W \times W$, $\approx = \{\{w\}, \{u\}\}$, $V(p) = \{w\}$;
- $N_1(w) = \{\{w, u\}\}$, $N_2(w) = \{\{w\}, \{w, u\}\}$, $N_1(u) = N_2(u) = \{W\}$.

It is not hard to see that $M_1, w \models [s:p!]\mathbb{O}_s\bot$ and $M_2, w \not\models [s:p!]\mathbb{O}_s\bot$. However, by an induction on the structure of ψ, we can show that $M_1, y \models \psi$ iff $M_2, y \models \psi$ for all $\psi \in \mathcal{L}_0$ and $y \in W$. Here we show only the inductive step for $\mathbb{O}_s\chi$. The

case for $y = u$ follows directly from that $N_1(u) = N_2(u)$ and the IH. For $y = w$, we have:

$$\begin{array}{lll} & M_1, w \models \mathbb{O}_s\chi & \\ \text{iff} & \forall U \in N_1(w),\, U \subseteq\sim(w) \text{ implies } U \subseteq [\![\chi]\!]_{M_1} & \text{(semantics)} \\ \text{iff} & \forall U \in N_1(w),\, U \subseteq [\![\chi]\!]_{M_1} & \text{(def. of } N_1(w) \text{ and } \sim) \\ \text{iff} & \forall U \in N_1(w),\, U \subseteq [\![\chi]\!]_{M_2} & \text{(IH)} \\ \text{iff} & \forall U \in N_2(w),\, U \subseteq [\![\chi]\!]_{M_2} & (\bigcup N_1(w) = \bigcup N_2(w)) \\ \text{iff} & \forall U \in N_2(w),\, U \subseteq\sim(w) \text{ implies } U \subseteq [\![\chi]\!]_{M_2} & \text{(def. of } N_2(w) \text{ and } \sim) \\ \text{iff} & M_2, w \models \mathbb{O}_s\chi & \text{(semantics)} \end{array}$$

(2): (2) can be shown similarly to (1). To obtain proof, we replace the formula $[s\!:\!p!]\mathbb{O}_s\bot$ in the proof of (1) by $[r\!:\!p?]\mathbb{O}_s\bot$. □

The remainder of the section is devoted to the second question. We want to know whether the notion of the power to know characterized in LRK is reducible to (a combination of) other operators. To do this, we consider the following sublanguage of $\mathcal{L}$ without the operator $R_r\varphi$:

$$\varphi ::= p \mid \neg\varphi \mid (\varphi \to \varphi) \mid K_r\varphi \mid \mathbb{O}_s\varphi \mid \Box\varphi \mid [s\!:\!\varphi!]\varphi \mid [r\!:\!\varphi?]\varphi \qquad (\mathcal{L}_3)$$

The usual way to proceed is to find a suitable notion of "bisimulation" for $\mathcal{L}_3$. In the next definition, we define the "almost-identical" relation between two models. Although the notion is much stronger than the usual notion of bisimulation, it is sufficient for our current purpose.

Definition 4.2 Let $M = (W, \sim, \approx, N, V)$ and $M' = (W', \sim', \approx', N', V')$ be two models. We say M and M' are *almost-identical* if the following hold:

(A1) $W = W'$, $\sim = \sim'$, $N = N'$, $V = V'$, and

(A2) for all $w \in W$ and $U \in N(w)$, $U \subseteq\approx(w)$ and $U \subseteq\approx'(w)$.

Lemma 4.3 *For all models $M = (W, \sim, \approx, N, V)$ and $M' = (W', \sim', \approx', N', V')$, if M and M' are almost-identical then $M, w \models \varphi$ iff $M', w \models \varphi$ for all $w \in W$ and $\varphi \in \mathcal{L}_3$.*

Proof. Induction on the structure of φ. We show only the cases for $[s\!:\!\psi!]\chi$ and $[r\!:\!\psi?]\chi$ since the remaining are all straightforward.

Case $[s\!:\!\psi!]\chi$: we have

$$\begin{array}{lll} & M, w \models [s\!:\!\psi!]\chi & \\ \text{iff} & M, w \not\models \psi \text{ or } M_{\psi!}, w \models \chi & \text{(semantics)} \\ \text{iff} & M', w \not\models \psi \text{ or } M_{\psi!}, w \models \chi & \text{(IH)} \\ \text{iff} & M', w \not\models \psi \text{ or } M'_{\psi!}, w \models \chi & \text{(IH)} \\ \text{iff} & M', w \models [s\!:\!\psi!]\chi & \text{(semantics)} \end{array}$$

Note that the third "iff" holds because $M_{\psi!}$ and $M'_{\psi!}$ are almost-identical since $[\![\psi]\!]_M = [\![\psi]\!]_{M'}$ by the IH.

Case $[r\!:\!\psi?]\chi$: we have [9]

[9] We need to slightly modify the definition of $M_{\psi?}$ (and $M'_{\psi?}$) below since the operator $R_r\psi$

	$M, w \models [r : \psi?]\chi$	
iff	$M_{\psi?}, w \models \chi$	(semantics)
iff	$M, w \models \chi$	($M_{\psi?} = M$ by the condition (A2))
iff	$M', w \models \chi$	(IH)
iff	$M'_{\psi?}, w \models \chi$	($M'_{\psi?} = M'$ by the condition (A2))
iff	$M', w \models [r : \psi?]\chi$	(semantics)

The next proposition gives a negative answer to the second question. The operator $R_r\varphi$ does express an independent notion.

Proposition 4.4 *There is no formula $\varphi \in \mathcal{L}_3$ such that $\varphi \equiv R_r p$.*

Proof. Consider two models $M = (W, \sim, \approx, N, V)$ and $M' = (W, \sim, \approx', N, V)$ such that $W = \{w, v\}$, $\sim = W \times W$, $N(w) = \{\{w\}\}$, $N(v) = \{\{v\}\}$, $V(p) = \{w\}$, and

- $\approx = \{\{w\}, \{v\}\}$,
- $\approx' = \{\{w, v\}\}$.

It is easy to see that $M, w \models R_r p$ and $M', w \not\models R_r p$. However, note that M and M' are almost-identical. Hence, $M, w \models \varphi$ iff $M', w \models \varphi$ for all $\varphi \in \mathcal{L}_3$ by Lemma 4.3. □

Corollary 4.5 *$\mathcal{L}$ is more expressive than $\mathcal{L}_3$.*

5 Related Work

Deontic logic for epistemic actions. In general, LRK is a specific deontic logic for actions in the epistemic context. Several attempts at developing deontic logic for epistemic actions can be found in the literature, e.g., [2], [1], and [12].

A logic for the notion of permitted announcements has been developed in [12]. The logic is based on the idea that a piece of information φ is permitted to be announced ($\mathbb{P}\varphi$) if the *epistemic state* after the announcement is ideal. The notion of epistemic state can be understood either as a syntactic notion or as a semantic notion. Syntactically, an epistemic state is just a set of epistemic formulas representing the knowledge of an agent. In contrast, from the semantic perspective, an epistemic state is a set of indistinguishable possible worlds by an agent (or, equivalently, an indistinguishability relation over all possible worlds). In [12], two semantic definitions of permitted announcements have been proposed based on the two understandings of epistemic states. Interestingly, the two definitions are shown to be equivalent, in the sense that they give the same set of logical validities. In this paper, we consider epistemic states as a semantic notion and apply the "neighbourhood epistemic model" introduced in [12]. Formally, a neighbourhood epistemic model is a structure $M = (W, \sim, N, V)$ where W, $\sim$, and V are the same as in the standard epistemic models (S5 models) and $N : W \to \wp(\wp(W))$ is a neighbourhood function assigning a set of *ideal* epistemic states to each possible world. Note that the

does not appear in $\mathcal{L}_3$. For example, $M_{\psi?}$ should be defined as "$M_{\psi?} = M$ if it is not the case that for all $U \in \approx$, $U \subseteq [\![\psi]\!]_M$ or $U \subseteq [\![\neg\psi]\!]_M$; otherwise ...".

public announcement of a proposition φ may restrict the current epistemic state to only those possible worlds satisfying φ. Hence, in [12], the formula $\mathbb{P}\varphi$ is interpreted in such a way that it is true at a possible world w iff the epistemic state after the announcement of φ is an element of $N(w)$. In this paper, we employ a similar idea to provide semantics for the notion of obligatory announcements.

In [2], two binary operators $P(\psi, \varphi)$ and $O(\psi, \varphi)$ are introduced to express the notions that "after announcing ψ, it is permitted/obligatory to announce φ". It is clear that our operator $\mathbb{O}_s\varphi$ can be expressed in their framework as $O(\top, \varphi)$. Conversely, the operator $O(\psi, \varphi)$ can be expressed as $[s : \psi!]\mathbb{O}_s\varphi$ in LRK. In [2], a ternary relation $\mathcal{P} \subseteq S \times \wp(S) \times \wp(S)$ is used to provide the semantics for $O(\psi, \varphi)$ in such a way that $\mathcal{M}, s \models O(\psi, \varphi)$ iff for all $(s, [\![\psi]\!]_M, S'') \in \mathcal{P}, S'' \subseteq [\![\langle\psi\rangle\varphi]\!]_M$, where S is the domain of the model. The major difference between the semantics for $\mathbb{O}_s\varphi$ and $O(\psi, \varphi)$ is that our operator $\mathbb{O}_s\varphi$ is specific to the receiver's knowledge. This is reflected in the fact that the formula $K_r(\varphi \to \psi) \to (\mathbb{O}_s\varphi \to \mathbb{O}_s\psi)$ is valid in LRK whereas not in the logic of [2]. We think that LRK is more suitable for reasoning about obligatory announcements in the context of, e.g., database security, because the receiver's initial knowledge is crucial for the sender's decision on which information should be disclosed, as suggested by Example 2.7.

An alternative definition of obligatory announcements has also been proposed in [1, Definition 10]. Aucher et al. [1] define the obligatory message of a security monitor as the minimal informative message such that, by sending it, the privacy policy compliance is restored. Clearly, the notion of "obligatory message" is different from the notion of obligatory announcements in our paper and [2]. But a detailed conceptual analysis of the difference between these notions is beyond the scope of the paper.

Logic of questions. The semantics of questions or interrogatives has received much attention in logic, see [7]. The basic idea is that the meaning of a question is what counts as an answer to that question. For example, the question "Is it raining in Guangzhou?" has two possible answers: "It is raining in Guangzhou" and "It is not raining in Guangzhou". Observe that they are both propositions or statements and, furthermore, they logically exclude each other and are jointly exhaustive. Hence, some logicians propose that questions can be represented semantically as a partition over the set of all possible worlds (or the logical space), e.g., [7], [4], and [21]. The semantics for questions proposed in [7], [4], and [21] are different. In this paper, we follow the approach in [21], because it is a conservative extension to the standard epistemic logic and we want to focus on the semantics for the power to know.

In [21], the so-called *epistemic issue models* are used to provide the semantics for questions. Formally, they are structures $M = (W, \sim, \approx, V)$ where the only novel thing is the equivalence relation $\approx$ on W (or, equivalently, a partition of W). Instead of having a single modality expressing that "the question whether φ is one of the current issues", a complex formula is used in [21] to express the notion. Technically, in addition to the modality $K\varphi$ for knowledge,

there are two new modalities $U\varphi$ and $Q\varphi$ where $U\varphi$ is the familiar universal modality and the truth definition of $Q\varphi$ is as follows:

$$M, w \models Q\varphi \text{ iff for all } v \in W\text{: } w \approx v \text{ implies } M, v \models \varphi$$

The notion that "the question whether φ is one of the current issues" is then expressed as the formula $U(Q\varphi \vee Q\neg\varphi)$.

As mentioned above, the idea behind the logic of question in [21] is that a set of questions can be represented as a partition over the logical space. But the logic remains neutral about where the partition or the set of questions is induced. For example, it can be induced either by a conversation, or by a game, or even by a research program [21]. In this paper, we assume that the partition is induced by the part of a normative system, such as a privacy policy, stipulating the questions an agent has the power to know the answers to.

Our work is closely related to [21] as the fragment of LRK on the operator $R_r\varphi$ is roughly a reinterpretation of the static logic $\mathrm{EL_Q}$ in [21]. However, there is also an important difference between our paper and [21]. In [21], there is also a dynamic operator $[\varphi?]\psi$ expressing that ψ holds after asking the question whether φ. But the effect of $[\varphi?]$ is to add the question $\varphi?$ to the set of current issues. In contrast, the operator $[r:\varphi?]\psi$ in LRK captures the deontic aspect of asking questions. It seems more appropriate to interpret the operator $[r:\varphi?]$ in our paper as the action that *commands* the sender to inform whether φ.

Logic of legal competence. The right to know is an epistemic right, and thus is a form of right. Works on the logical analysis of legal rights can also be found in the literature, e.g., [11,13,14,10]. Recent works on the topic investigating explicitly the power type of right includes [15,19,6]. Given that these works are on general (power-)rights, one may wonder why there is a need to develop a separate logic for epistemic rights. One reason is that there are some valid reasoning patterns for epistemic rights that can not be expressed in a language devised for general rights, just like not all principles of public announcement logic can be expressed in dynamic logic.

In our paper, we treat the power to know as an independent notion. As pointed out by [6], there exist two different logical approaches formalizing legal power: an earlier tradition reduces power to (a combination of) obligations, permissions, and actions ([11,13]); whereas the other (e.g.[14,10]) holds that power is not reducible to static normative positions, which follows the original separation of Hohfeld (see [15]). Thus, our work adopts the second approach. We also show, in Proposition 4.4, that the notion of the power to know characterized in LRK can not be expressed in a language without it.

6 Conclusion and Future Work

In this paper, a logic LRK was introduced semantically for reasoning about the power to know, the obligatory announcements, and the dynamics of questions and public announcements. We explored some (in)validities of LRK, where the interaction between the power to know and the obligatory announcements has been highlighted. We also studied the expressive power of several fragments of

the language of LRK. We showed that the incorporation of the two dynamic operators in LRK increases the expressive power and the operator for the power to know cannot be expressed in a language without it.

There are many directions for future research. A natural task is to investigate some metalogical properties of LRK, such as axiomatization, completeness, and computational complexity. We can also consider extensions to LRK. For example, it is interesting to reason about the ability of the receiver in LRK since the receiver may use their power to know by asking (a sequence of) questions. Other interesting extensions to LRK include the incorporation of "the power not to know" [16], how to model the actions of adding or removing the receiver's power to know, how to extend to the multi-agent case, etc.

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